James Byrne

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Education

University of Waterloo - BASc Electrical Engineering (Honours, Co-op), 4B, Expected Graduation: April 2027

Skills

Languages: C++, C, Python, VHDL
Embedded & Systems: RTOS, FreeRTOS, STM32, ARM Cortex-M, POSIX API, BSP, VFS, Device Drivers
Protocols & Peripherals: CAN, MQTT, I2C, SPI, UART
Tools & Development: VS Code, Make, OpenOCD, GDB, STM32CubeIDE, Docker, KiCad, Git, MATLAB

Work Experience

Apple - SoC Embedded Firmware Intern - Cupertino, CA — September 2026 - December 2026

- Contributed to an RTOS used for SoC testing and validation, developing new features and fixing bugs.
- Designed and implemented a firmware library and console command set to power up, load, boot, control and monitor an auxiliary core from the main RTOS.
- Authored multiple bare-metal firmware programs to be loaded and run on the auxiliary core at runtime.
- Developed a background monitoring task in the main RTOS to receive logs from the auxiliary core using interrupt-driven event handling and single-writer shared memory.
- Implemented fault capture and status decoding for the auxiliary core, integrated crash reports into the main RTOS crash dump path and validated functionality using fault injection firmware.
- Built a Python and Make pipeline that processes the auxiliary core's firmware build artifacts to generate the headers and packed images the main RTOS uses to load the firmware at runtime.
- Added large block mapping support to the main RTOS's system MMU driver for DMA agents, mapping memory regions with the largest aligned blocks instead of only the smallest page size.

AlumaPower Corporation - Firmware Engineering Intern - Kitchener, ON — January 2026 - April 2026

- Implemented SNTP-based time synchronization using a Chrony server, modifying FreeRTOS POSIX layers and W6100 network drivers to provide system-wide POSIX timekeeping.
- Wrote POSIX ioctl drivers with VFS integration for MCF8316C, INA745, TCA9535, TCA9548 and MCP9843 chips.
- Contributed OS-level enhancements within the BSP layer, implementing VFS path construction, service threads, PHY link handling and resolving system stability bugs.
- Developed and debugged application-layer firmware across multiple distributed subsystems.
- Developed Node-RED dashboards and flows for MQTT telemetry visualization and subsystem control.

AlumaPower Corporation - Firmware Engineering Intern - Kitchener, ON — January 2025 - April 2025

- Developed and debugged embedded C firmware on STM32G474RE using Makefiles and OpenOCD, implementing FreeRTOS state machines for motor controller subsystems with MCF8316C controllers.
- Developed POSIX-style I2C drivers and control wrappers for the MCF8316C, including EEPROM abstraction layers.
- Contributed to BSP abstractions for flexible peripheral configuration (I2C, CAN, EEPROM) across subsystems.
- Created Python tools to automate testing, telemetry and logging for CAN/MQTT-based subsystems.
- Designed and assembled a remote debugging PCB with Raspberry Pi CM4 and STLINK-V3MODS for GDB/OpenOCD access without direct hardware connection.

AlumaPower Corporation - Systems Technician - Sarnia, ON — May 2024 - August 2024

- Developed FreeRTOS firmware in C for an STM32G474RE-controlled fluid management system.
- Implemented CAN TX/RX, interrupt-driven timers, a multi-state control state machine which included 1-Wire temperature sensors, level sensors and flow sensors for event triggering.
- Built a hardware safety shutdown circuit using DPDT relays driven by hydrogen, airflow and fluid level sensors.
- Created Python tools to filter and decode CAN messages sent over MQTT for real-time telemetry and debugging.

Ford Motor Company - R&D Test Engineering Intern - Windsor, ON — September 2023 - December 2023

- Conducted engine failure analysis and vibration testing using accelerometers and MATLAB.
- Managed engine dyno cells and developed VBA/SharePoint tools to automate calibration workflows.
- Designed and built an engine test bench using wiring harnesses to safely test individual engine components.

Clark Construction Management - Field Engineering Intern - Toronto, ON — January 2023 - April 2023

- Interpreted structural drawings to quantify materials and coordinate large-scale concrete pours (~6,400 m^3).

JC & J Byrne Farms - Ruscom Station, ON — 2015 - Present

- Maintain and troubleshoot mechanical and electrical equipment in agricultural operations.